Machine Learning 2

Super AI Engineer Season 6

Level 2 / 21 April 2026

Presented by

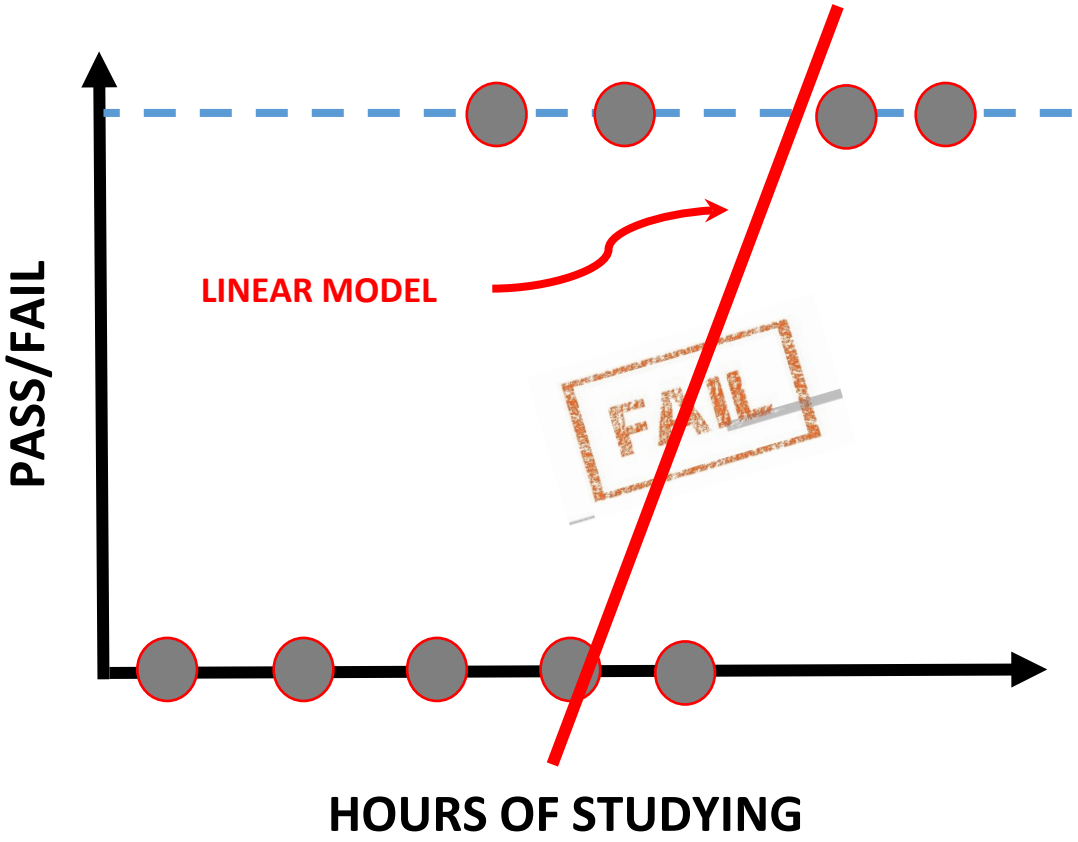
Aslan B. Wong

Arttie 可

LOGISTIC REGRESSION CLASSIFIER MODEL

LOGISTIC REGRESSION: INTUITION

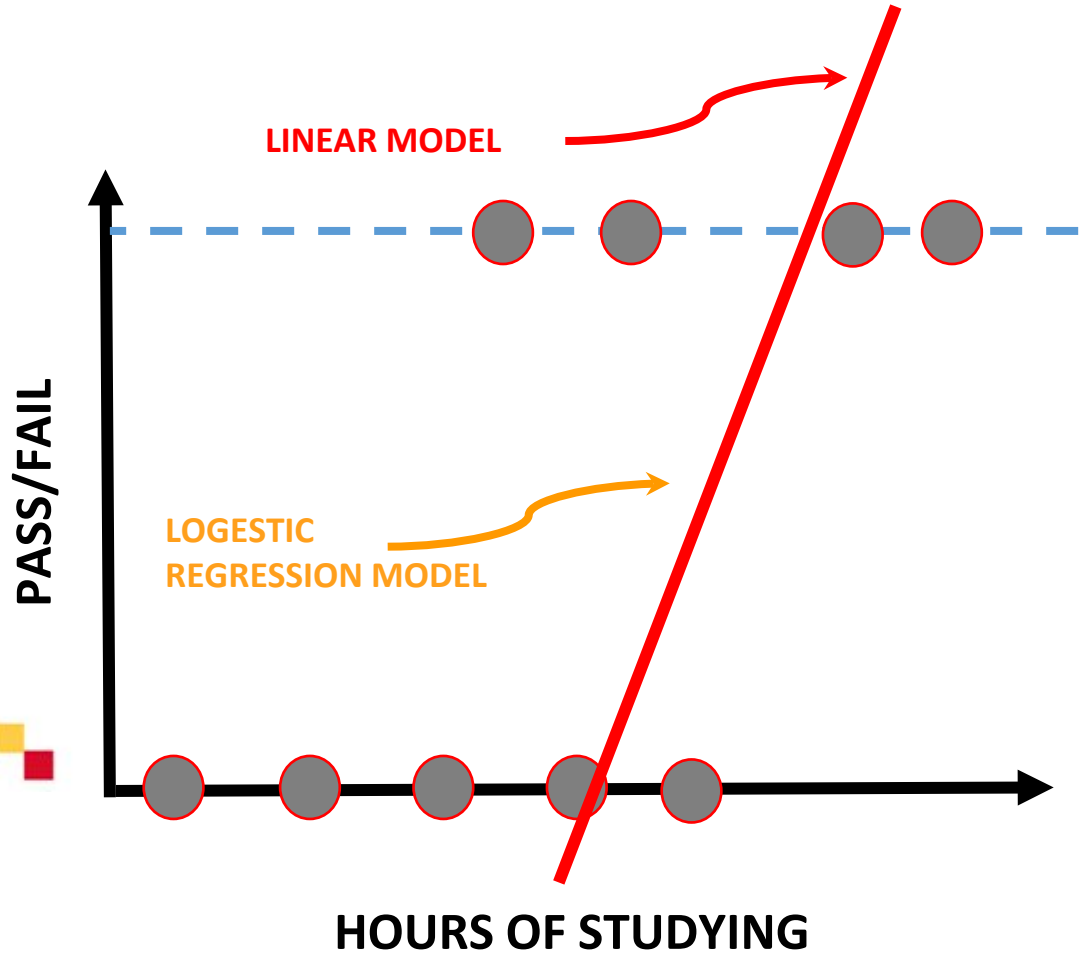
- Linear Regression is used to predict outputs on a continuous spectrum.
- Example: Predicting revenue based on the outside air temperature.
- Logistic Regression is used to predict binary outputs with 2 possible values (0 or 1).
- Example: Logistic model output can be one of two classes: pass/fail, win/lose, healthy/sick



Hours Studying	Pass/Fail
1	0
1.5	0
2	0
3	1
3.25	0
4	1
5	1

LOGISTIC REGRESSION: MATH

- Linear Regression is not suitable for classification problem.
- Linear Regression is unbounded, so Logistic Regression will be a better candidate in which the output value ranges from 0 to 1.



Linear Equation:

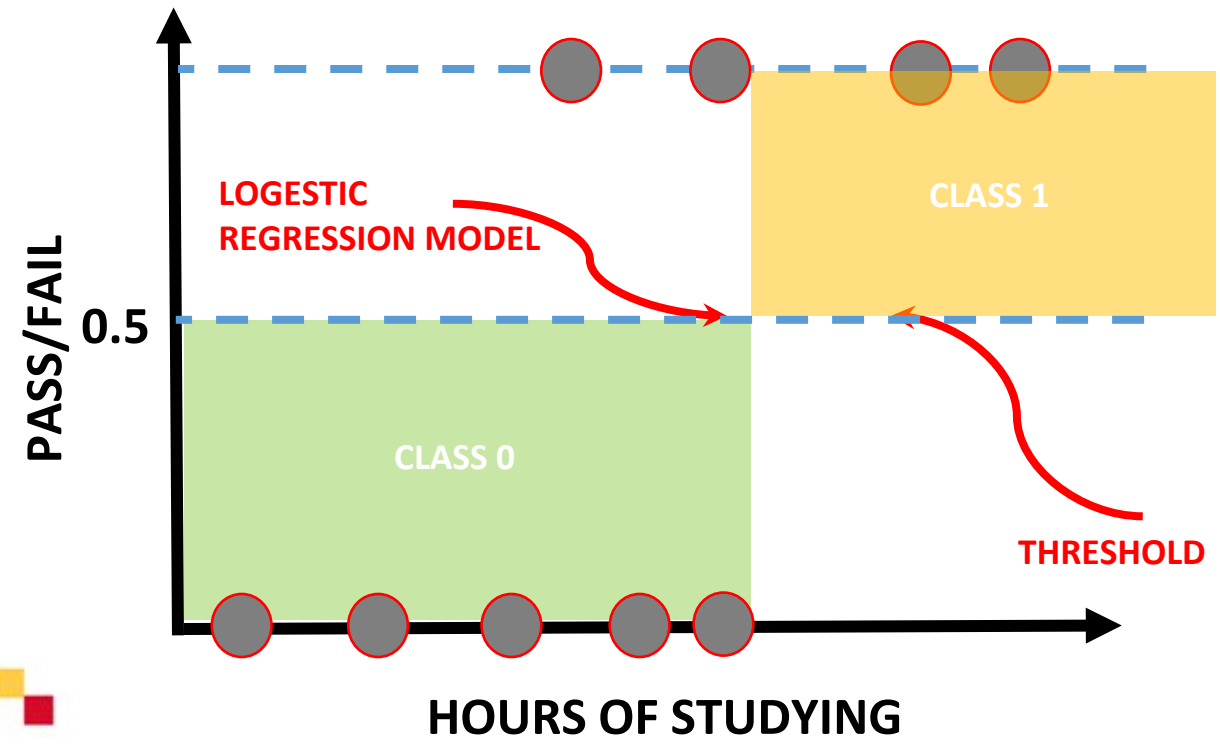
- $y = b_o + b_1 * x$

Apply Sigmoid Function:

- $P(x) = \text{sigmoid}(y)$
- $P(x) = 1/1+e^{-y}$
- $P(x) = 1/1+e^{-(b_0 + b_1 * x)}$

LOGISTIC REGRESSION: FROM PROBABILITY TO CLASS

- Now we need to convert from a probability to a class value which is “0” or “1”



Linear Equation:

- $y = b_o + b_1 * x$

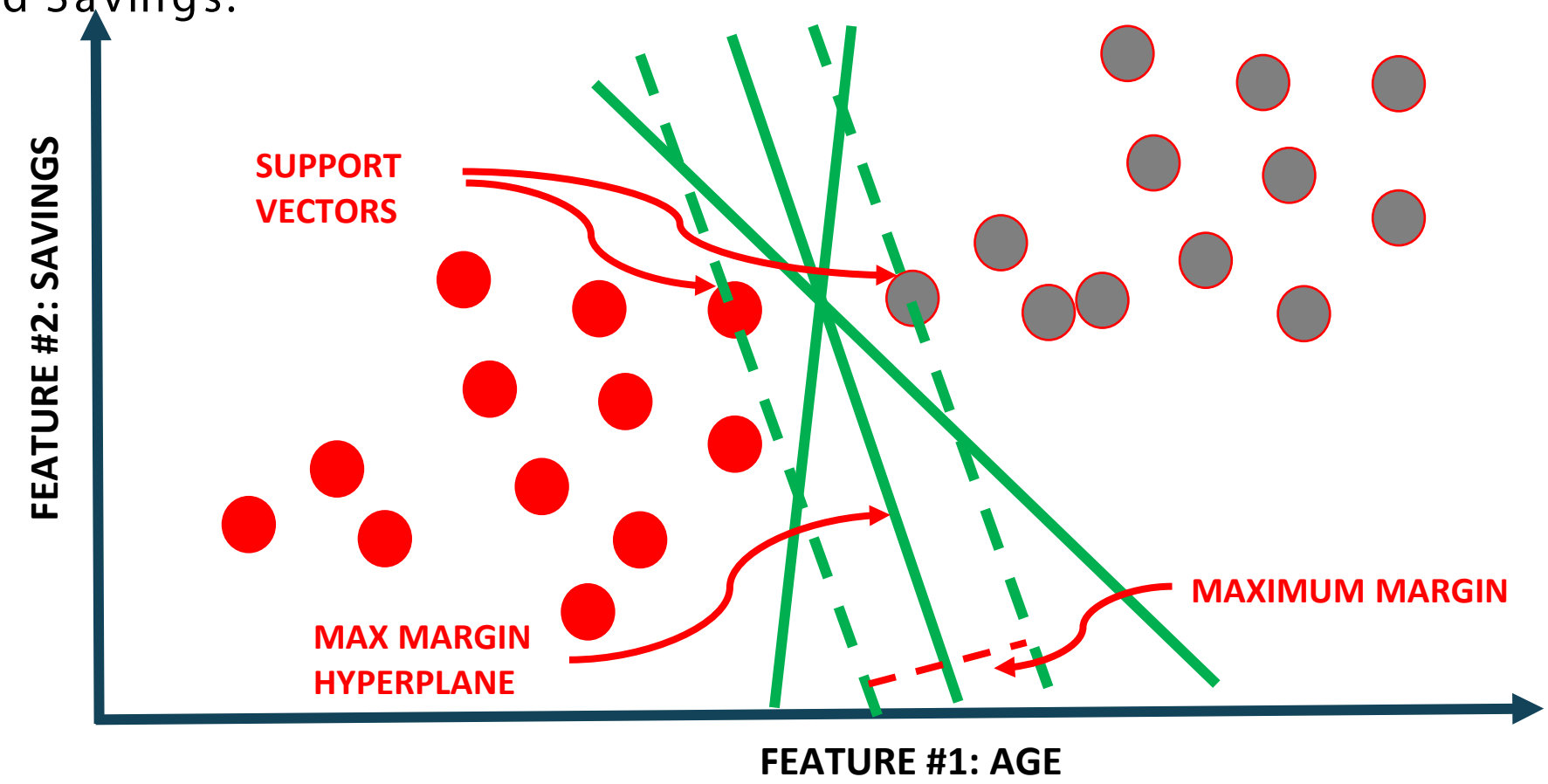
Apply Sigmoid Function:

- $P(x) = \text{sigmoid}(y)$
- $P(x) = 1/1+e^{-y}$
- $P(x) = 1/1+e^{-(b_0 + b_1 * x)}$

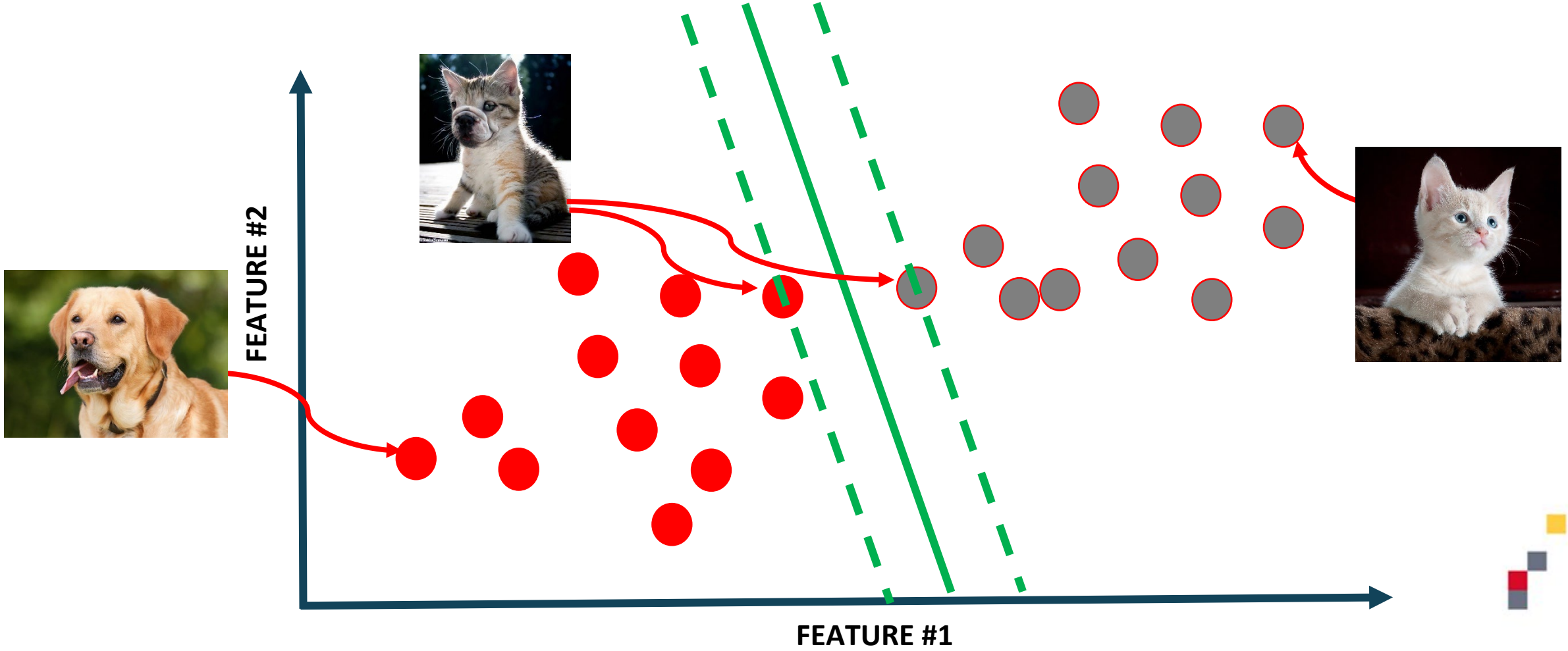
SUPPORT VECTOR MACHINES (SVM)

SUPPORT VECTOR MACHINES: INTUITION

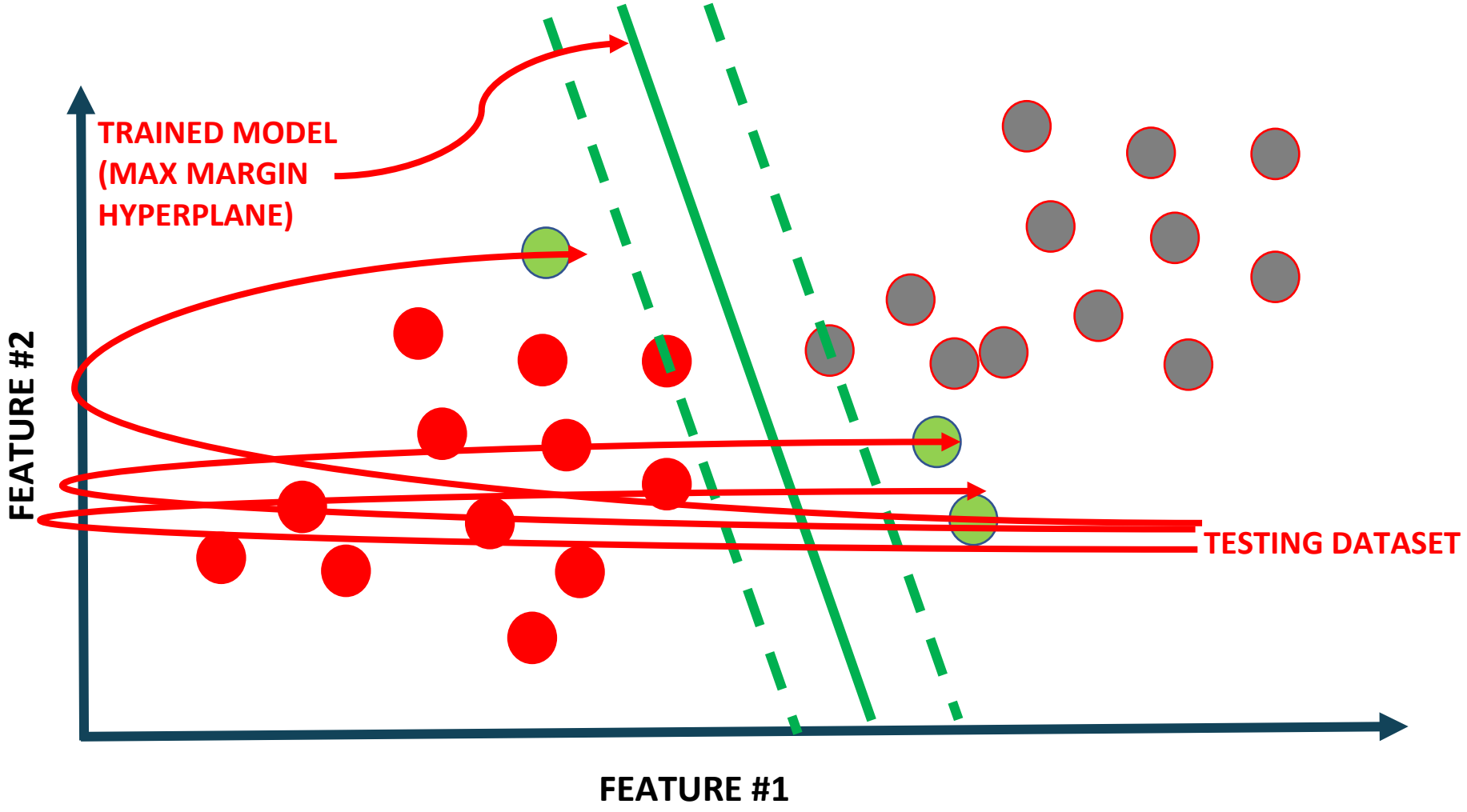
- Assume that you are data scientist working at a major bank in NYC.
- You want to classify a new client as eligible to retire or not, customer features are: Age and Savings.



SUPPORT VECTOR MACHINES: INTUITION

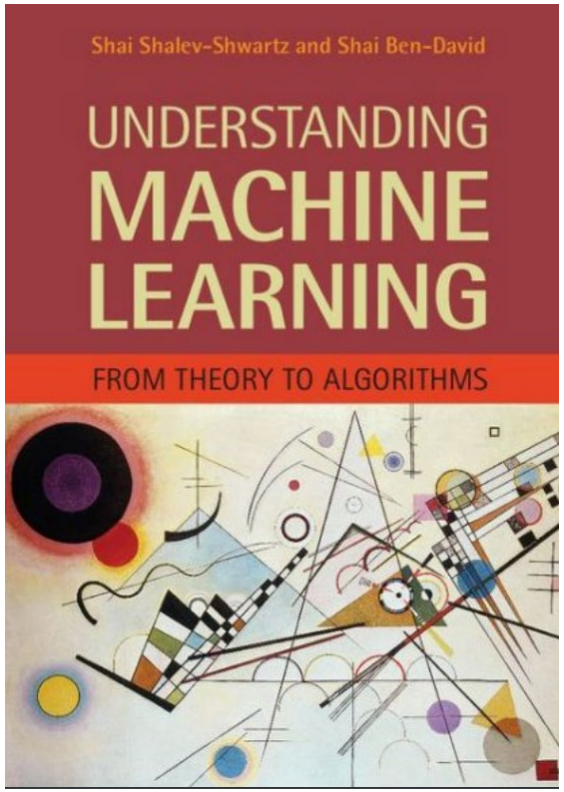


SUPPORT VECTOR MACHINES: MODEL EVALUATION

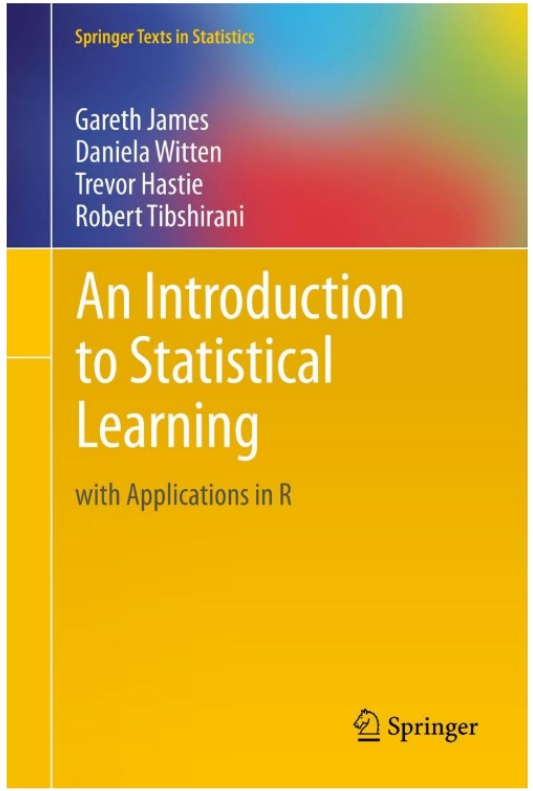


SUPPORT VECTOR MACHINES: ADDITIONAL READING MATERIAL

- Additional Resources, Page #202: http://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/understanding-machine-learning-theory-algorithms.pdf



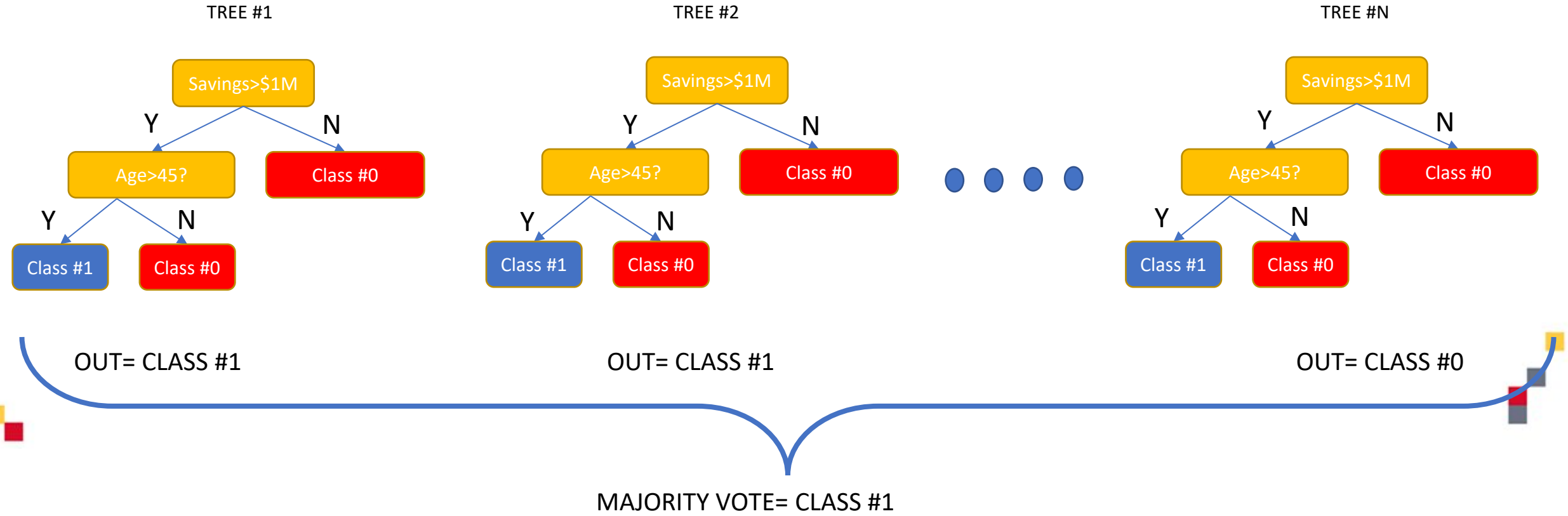
- Additional Resources, Page #337: An Introduction to Statistical Learning



RANDOM FOREST CLASSIFIER MODELS

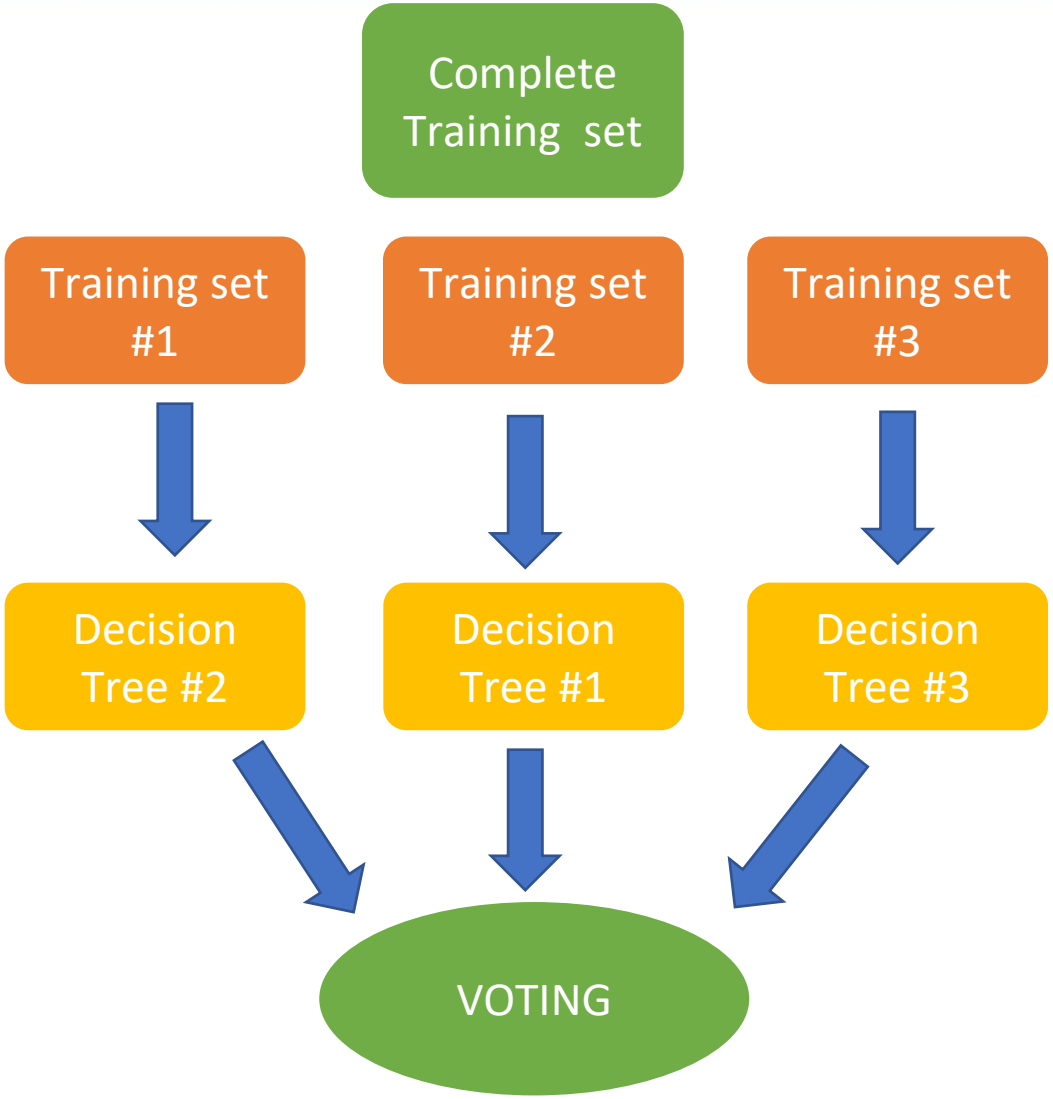
RANDOM FOREST CLASSIFIER: INTUITION

- Random Forest classifier is a type of ensemble algorithm
- It creates a set of decision trees from randomly selected subset of training set
- It then combines votes from different decision trees to decide the final class of the test object.



RANDOM FOREST: WHY AND HOW?

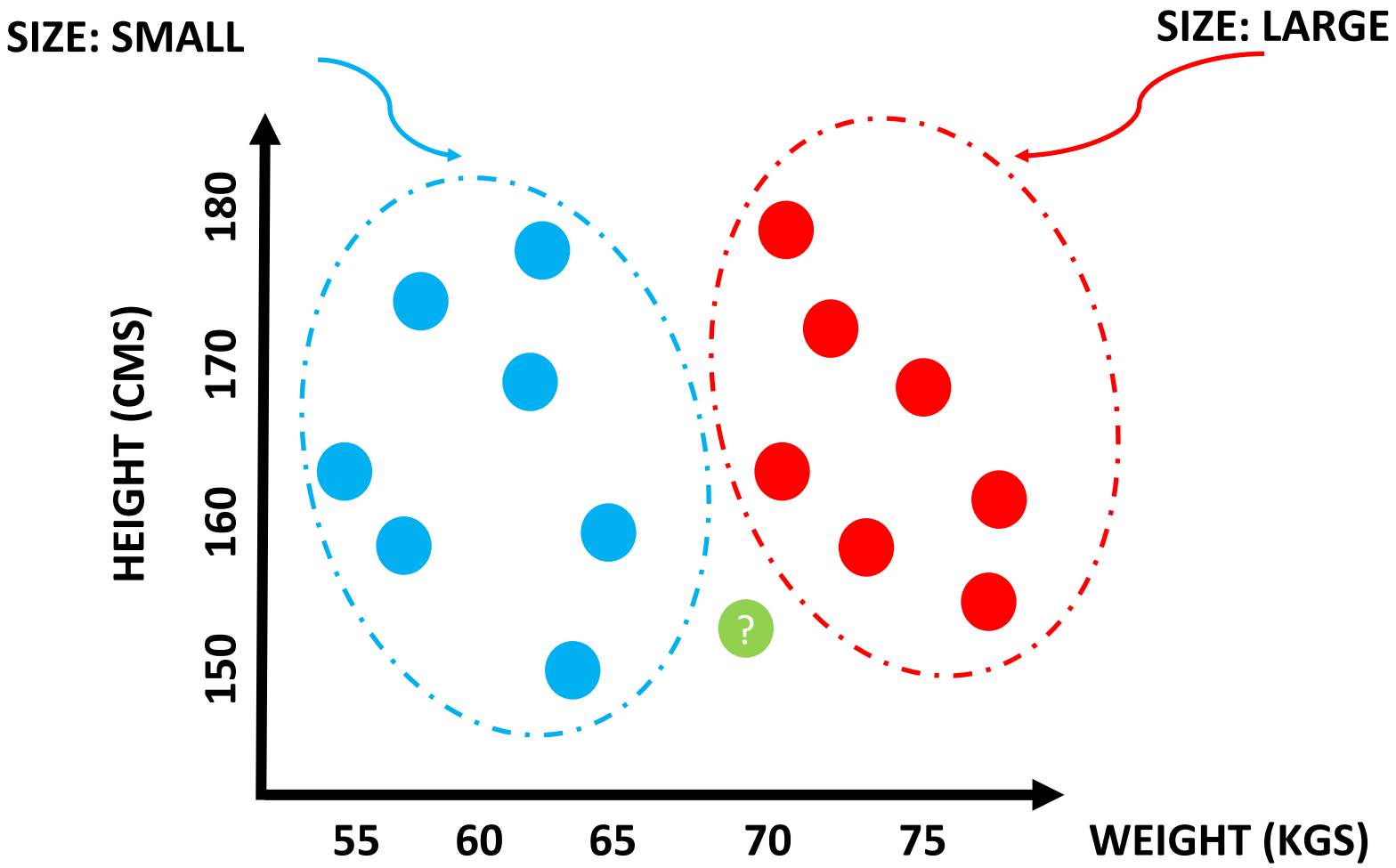
- It overcomes the issues with single decision trees by reducing the effect of noise
- Overcomes overfitting problem by taking average of all the predictions, canceling out biases
- Suppose training set: [X1, X2, X3, X4] with labels: [L1, L2, L3, L4]
- Random Forest creates three decision trees taking inputs as follows: [X1, X2, X3], [X1, X2, X4], [X2, X3, X4]
- Example: Combining votes from a pool of expert, each will bring their own experience and background to solve the problem resulting in a better outcome.
- Runs effectively on large database
- For large data, it produces highly accurate predictions



K-NEAREST NEIGHBOUR (KNN)

K NEAREST NEIGHBORS (KNN): INTUITION

- K-Nearest Neighbors (KNN) algorithm is a classification algorithm
- KNN works by finding the most similar data points in the training data, and attempt to make an educated guess based on their classifications



K NEARSET NEIGHBORS (KNN): ALGORITHM STEP

Select a value for k (e.g.: 1, 2, 3, …..)

→

Calculate the Euclidean distance between the point to be classified and every other point in the training data set

→

Pick the k closest data points (points with the k smallest distances)

→

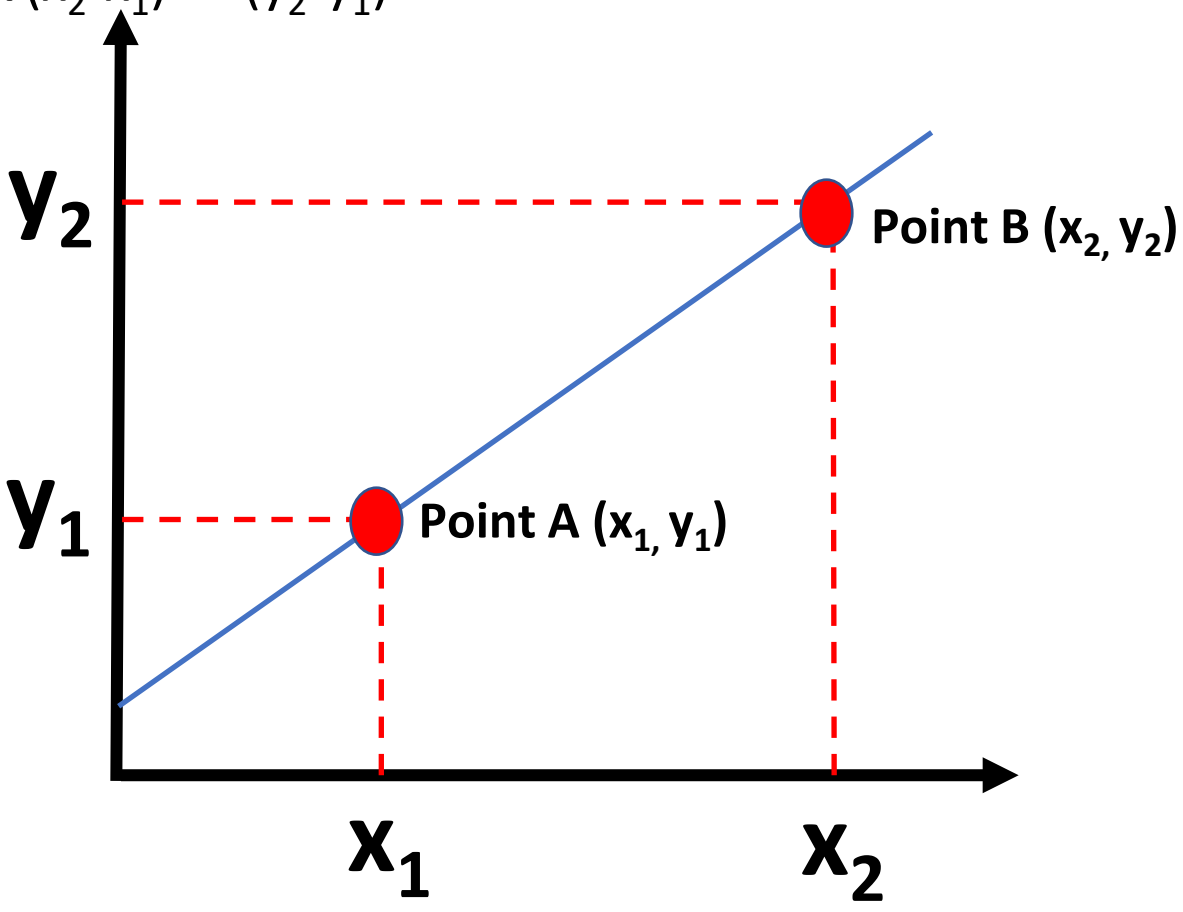
Run a majority vote among selected data points, the dominating classification is the winner. Point is classified based on the dominant class

→

Repeat

EUCLIDEAN DISTANCE: INTUITION

- Euclidean Distance= $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$



K NEARSET NEIGHBORS (KNN): EXAMPLE

- KNN will look for the 5 data points that are closest to the new customer data point
- The algorithm will determine which category (class) are these 5 points in
- Since 4 points had class "SMALL" & 1 had "LARGE", then new customer is assigned small size

Height	Weight	T-Shirt Size	Euclidian Dist	Vote
158	58	S	4.242640687	
158	59	S	3.605551275	
158	63	S	3.605551275	
160	59	S	2.236067977	3
160	60	S	1.414213562	1
163	60	S	2.236067977	3
163	61	S	2	2
160	64	L	3.16227766	5
163	64	L	4	
165	61	L	4.123105626	
165	62	L	5.656858249	

New Customer Information:

Height: 161
Weight: 61

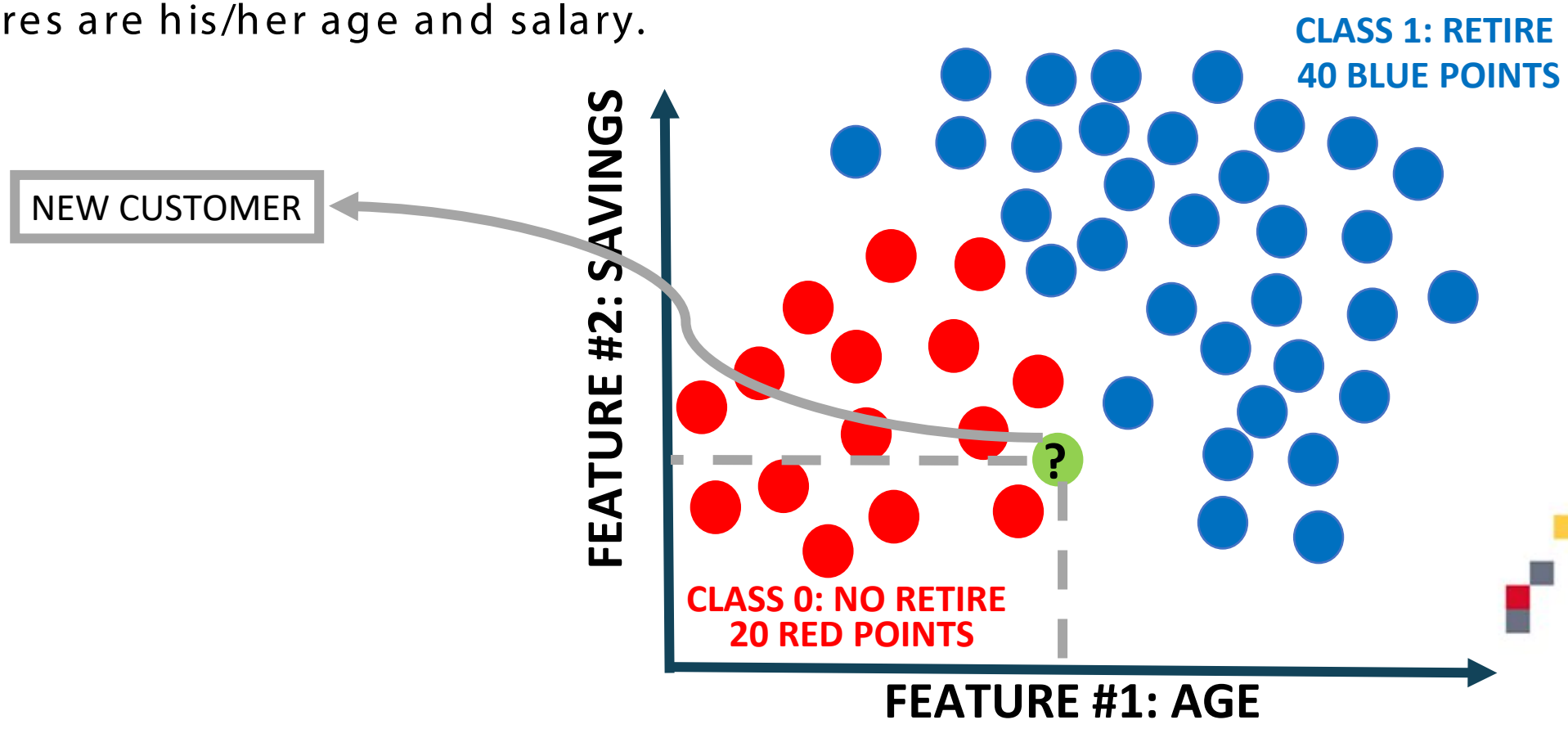
Assume, k= 5

Example Source: https://www.listendata.com/2017/12/k-nearest-neighbor-step-by-step-tutorial.html

NAÏVE BAYES CLASSIFIER MODEL

NAÏVE BAYES: INTUITION

- Naïve Bayes is a classification technique based on Bayes' Theorem.
- Let's assume that you are data scientist working major bank in NYC and you want to classify a new client as eligible to retire or not.
- Customer features are his/her age and salary.

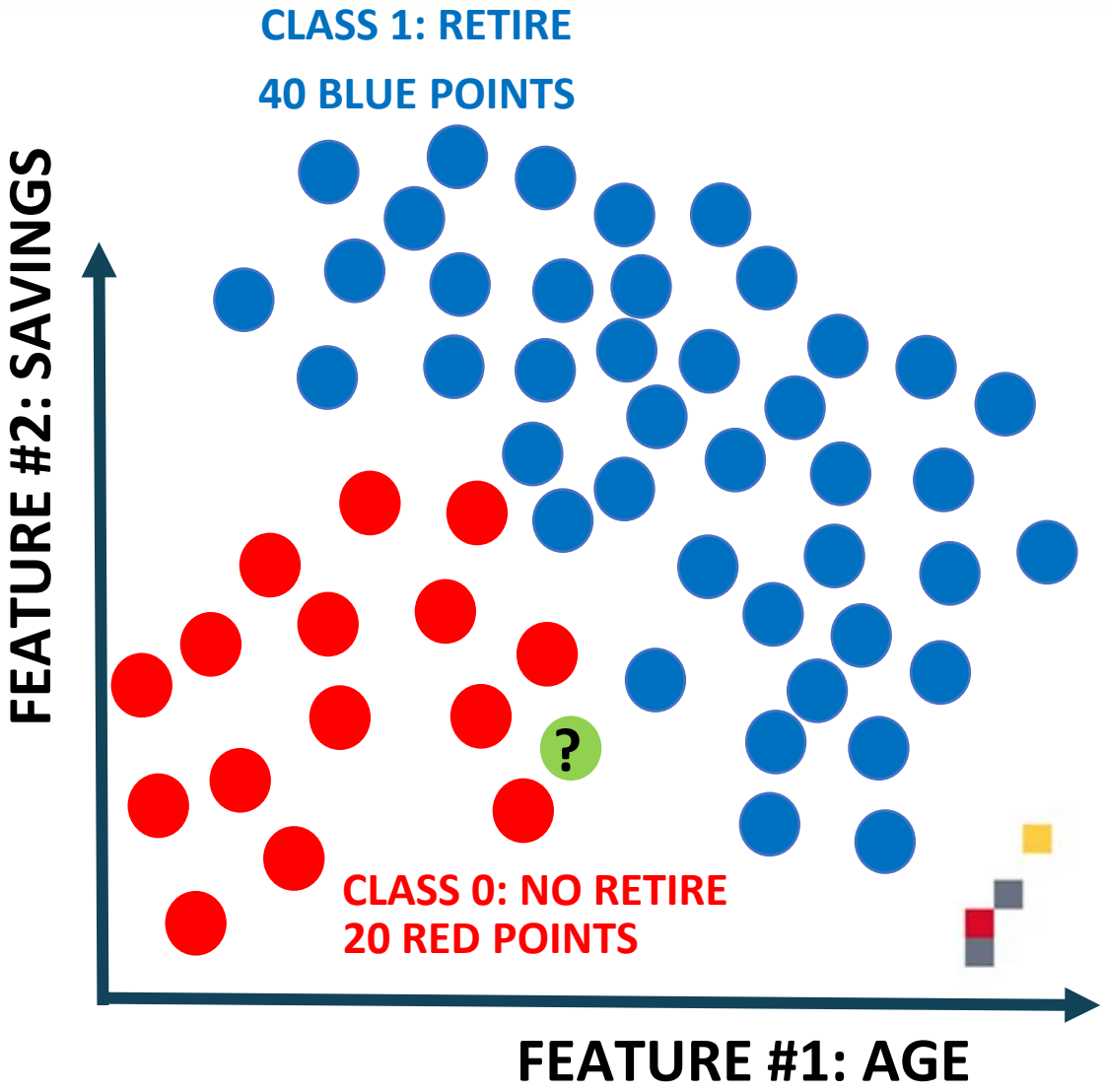


NAÏVE BAYES: 1. PRIOR PROBABILITY

- Points can be classified as RED or BLUE.
- Our task is to classify a new point to RED or BLUE.
- **Prior Probability:** Since we have more BLUE compared to RED, we can assume that our new point is twice as likely to be BLUE than RED.

$$Prior\ Probability\ for\ RED = \frac{Number\ of\ RED\ Points}{Total\ Number\ of\ Points} = \frac{20}{60}$$

$$Prior\ Probability\ for\ BLUE = \frac{Number\ of\ BLUE\ Points}{Total\ Number\ of\ Points} = \frac{40}{60}$$

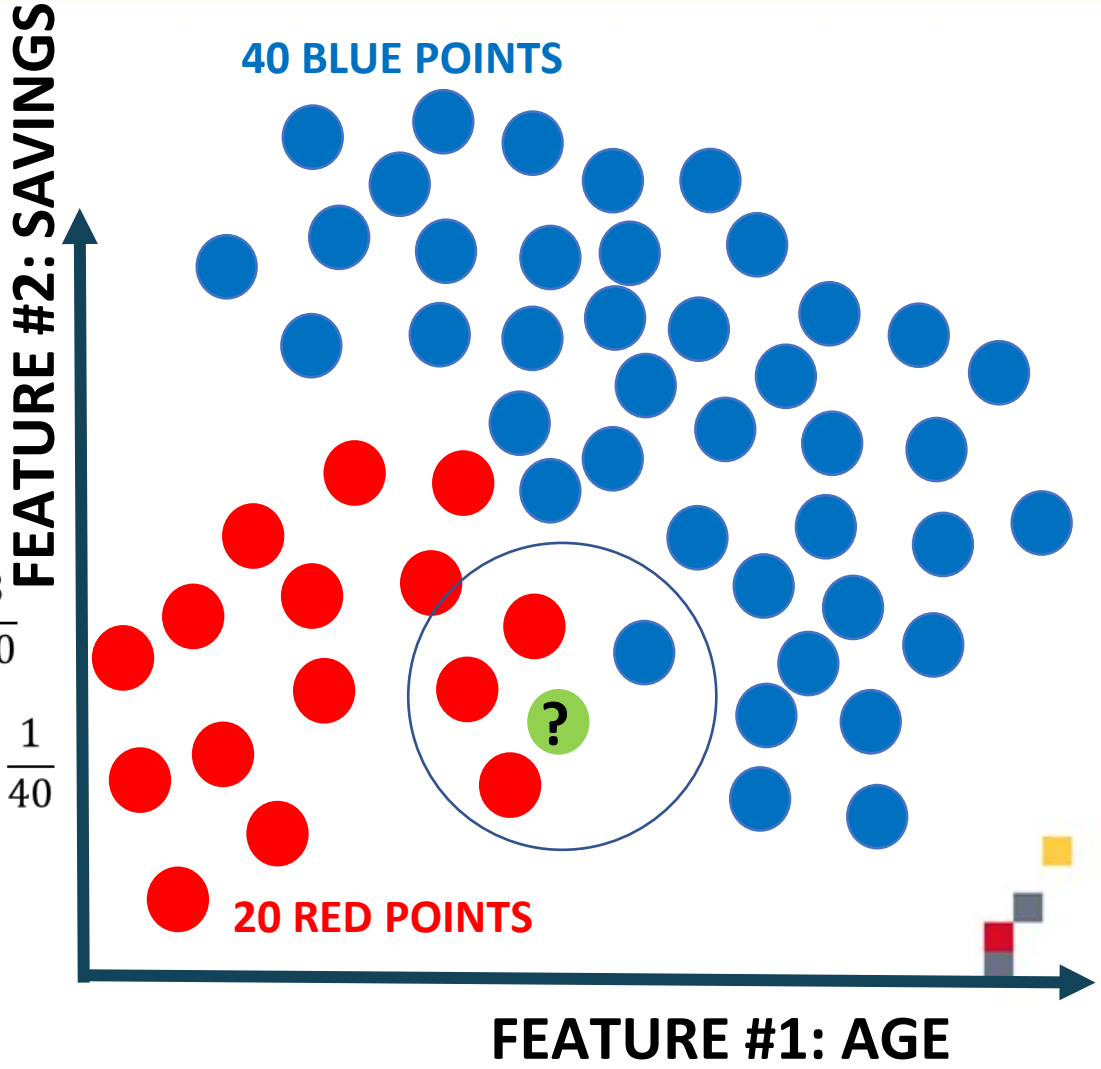


NAÏVE BAYES: 2. LIKELIHOOD

- For the new point, if there are more **BLUE** points in its vicinity, it is more likely that the new point will be classified as **BLUE**.
- So we draw a circle around the point
- Then we calculate the number of points in the circle belonging to each class label.

$$Likelihood\ of\ X\ being\ RED = \frac{Number\ of\ RED\ Points\ in\ vicinity}{Total\ Number\ of\ RED\ Points} = \frac{3}{20}$$

$$Likelihood\ of\ X\ being\ BLUE = \frac{Number\ of\ BLUE\ Points\ in\ vicinity}{Total\ Number\ of\ BLUE\ Points} = \frac{1}{40}$$

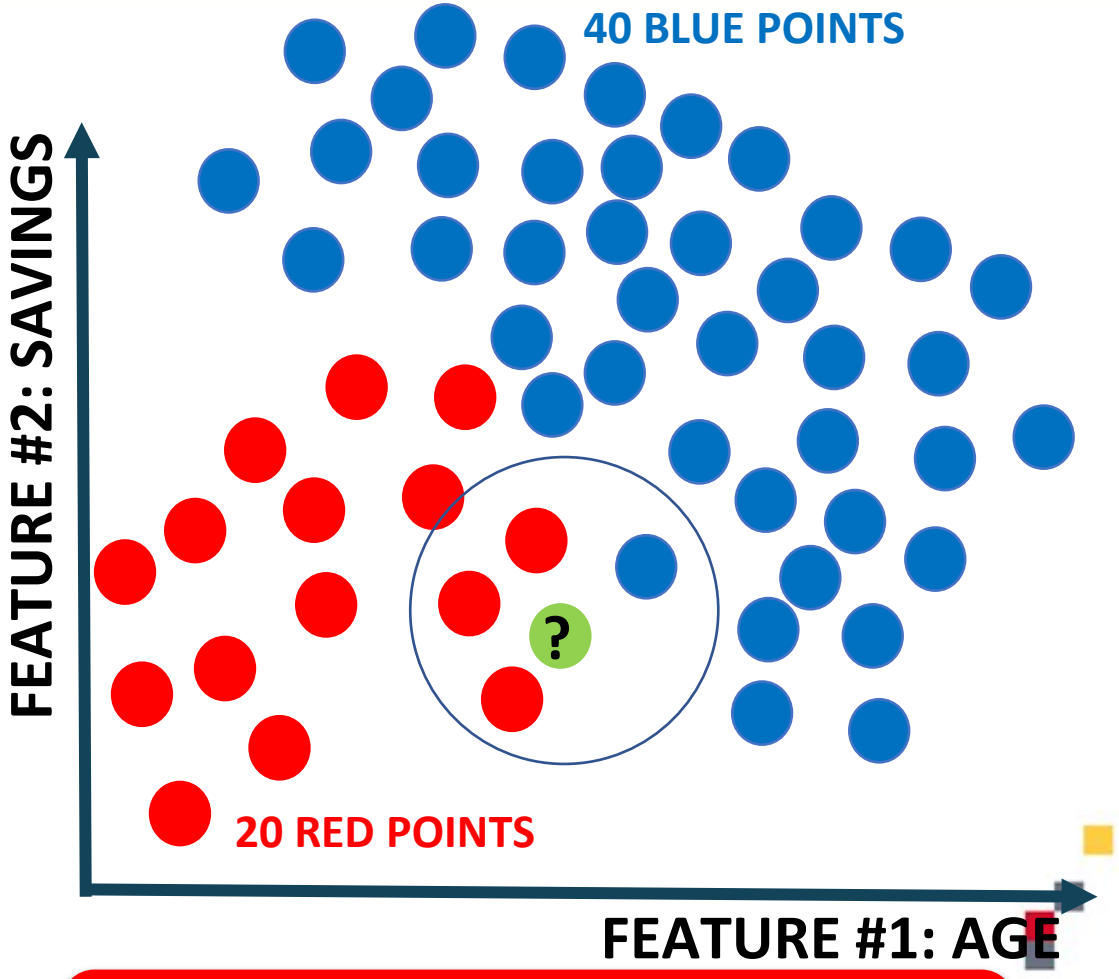


NAÏVE BAYES: 3. POSTERIOR PROBABILITY

- Let's combine prior probability and likelihood to create a posterior probability.
- **Prior probabilities:** suggests that X may be classified as BLUE Because there are twice as much blue points.
- **Likelihood:** suggests that X is RED because there are more RED points in the vicinity of X.
- Bayes' Rule combines both to form a posterior probability.

$$\begin{aligned}&Posterior\ Probability\ of\ X\ being\ RED\\&=\ Prior\ Probability\ of\ RED\ *\ Likelihood\ of\ X\ being\ RED\\&=\frac{20}{60}*\frac{3}{20}=\frac{1}{20}\end{aligned}$$

$$\begin{aligned}&Posterior\ Probability\ of\ X\ being\ BLUE\\&=\ Prior\ Probability\ of\ BLUE\ *\ Likelihood\ of\ X\ being\ BLUE\\&=\frac{40}{60}*\frac{1}{40}=\frac{1}{60}\end{aligned}$$



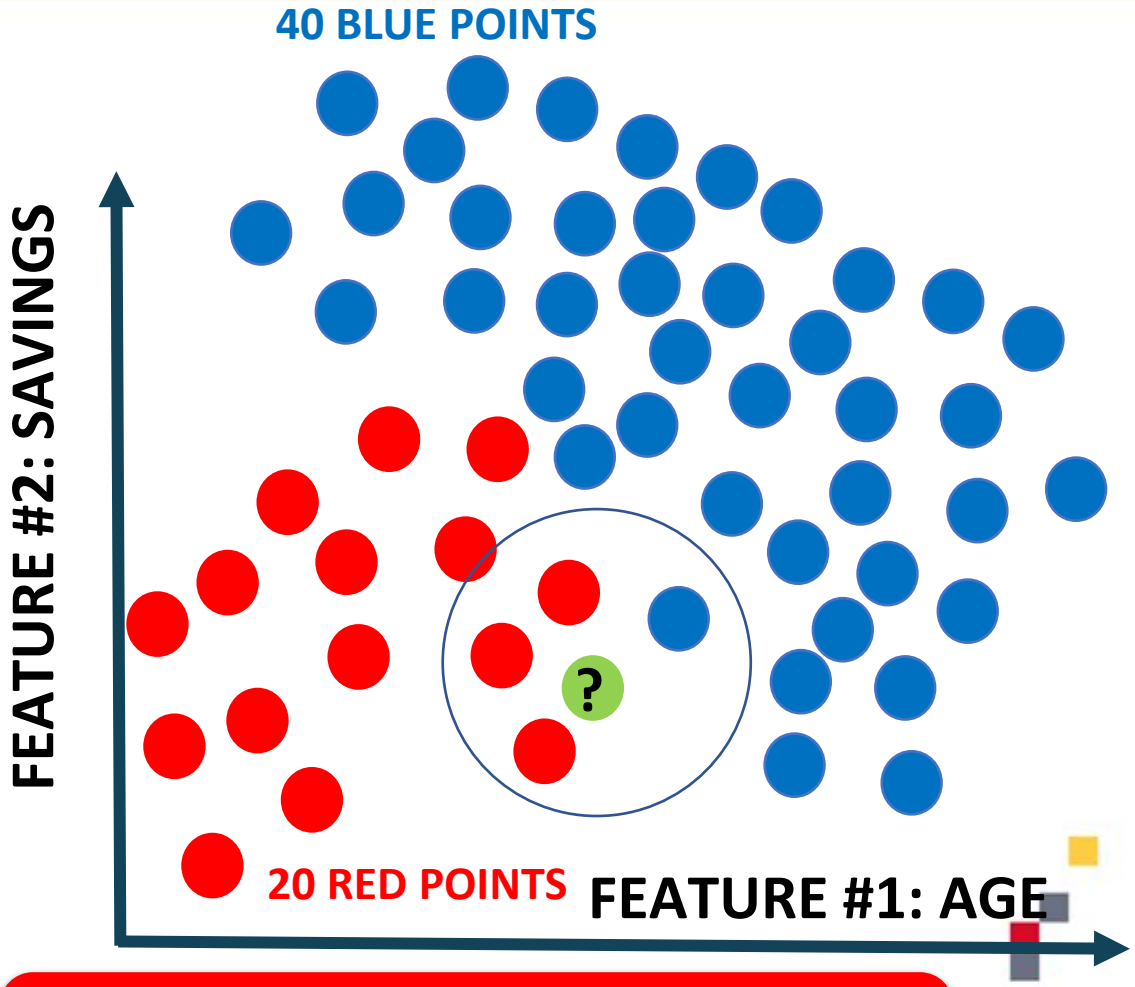
X CLASSIFIED AS RED (NON RETIRING) SINCE IT HAS LARGER POSTERIOR PROBABILITY

NAÏVE BAYES: REVIEW

- Let's combine prior probability and likelihood to create a posterior probability.
- **Prior probabilities:** suggests that X may be classified as BLUE Because there are twice as much blue points.
- **Likelihood:** suggests that X is RED because there are more RED points in the vicinity of X.
- Bayes' Rule combines both to form a posterior probability.

$$Posterior\ Probability\ of\ X\ being\ RED$$
$$= Prior\ Probability\ of\ RED * Likelihood\ of\ X\ being\ RED$$
$$= \frac{20}{60} * \frac{3}{20} = \frac{1}{20}$$

$$Posterior\ Probability\ of\ X\ being\ BLUE$$
$$= Prior\ Probability\ of\ BLUE * Likelihood\ of\ X\ being\ BLUE$$
$$= \frac{40}{60} * \frac{1}{40} = \frac{1}{60}$$



X CLASSIFIED AS RED (NON RETIRING) SINCE IT HAS LARGER POSTERIOR PROBABILITY

NAÏVE BAYES: SOME MATH!

- Naïve Bayes is a classification technique based on Bayes' Theorem.

LIKELIHOOD

PRIOR PROBABILITY OF RETIRING

$$P(Retire|X) = \frac{P(X|Retire) * P(Retire)}{P(X)}$$

MARGINAL LIKELIHOOD

- X: New Customer's features; age and savings
- $P(Retire|X)$: probability of customer retiring given his/her features, such as age and savings
- $P(Retire)$: Prior probability of retiring, without any prior knowledge
- $P(X|Retire)$: likelihood
- $P(X)$: Marginal likelihood, the probability of any point added lies into the circle

NAÏVE BAYES: SOME MATH!

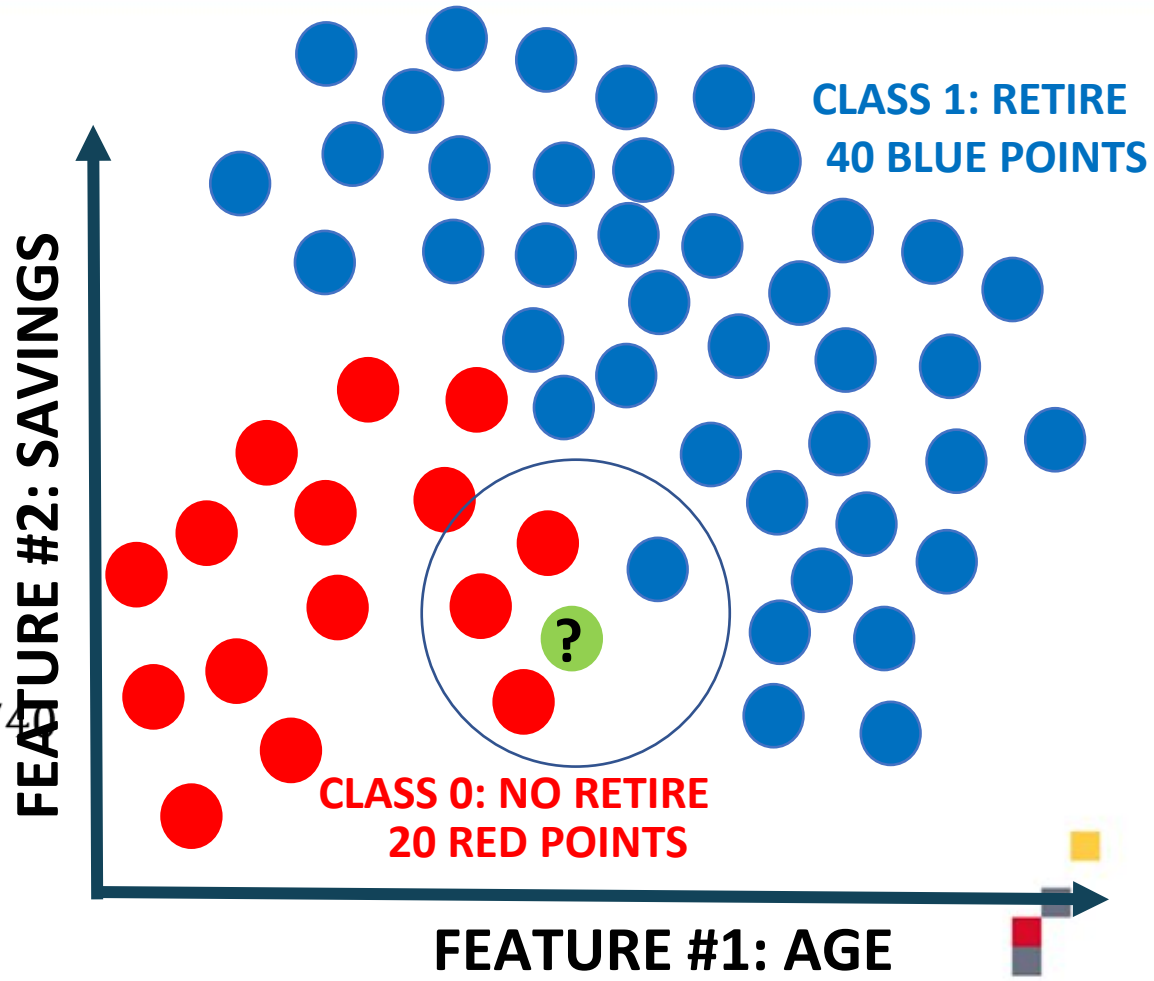
LIKELIHOOD

PRIOR PROBABILITY OF RETIRING

$$P(Retire|X) = \frac{P(X|Retire) * P(Retire)}{P(X)}$$

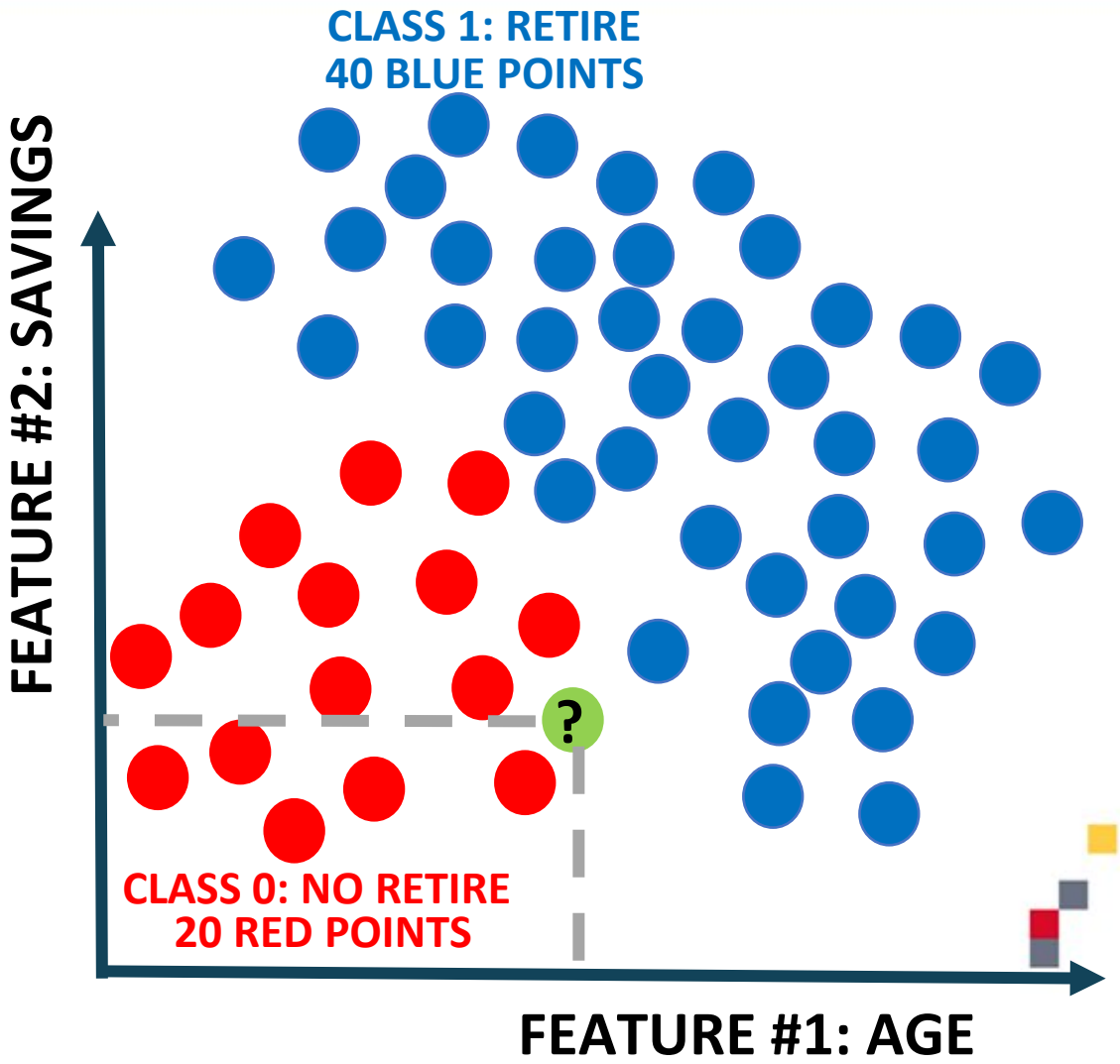
MARGINAL LIKELIHOOD

- $P(Retire) = \frac{\text{\# of Retiring}}{\text{Total points}} = 40/60$
- $P(X|Retire) = \frac{\text{\# of smilar observations for retiring}}{Total\ \#\ retiring} = 1/4$
- $P(X) = \frac{\text{\# of Similar observations}}{Total\ \#\ Points} = 4/60$
- $P(Retire|X) = \frac{\frac{40}{60} * \frac{1}{40}}{\frac{4}{60}} = \frac{1/60}{4/60} = 0.25$



NAÏVE BAYES: WHY NAÏVE?

- It is called naive because it assumes that the presence of a certain feature in a class is independent of the presence of other features.
- EXAMPLE #1: Age/savings, the assumption is not necessarily true since age and savings might be dependant on each others
- EXAMPLE #2: fruit can be classified as watermelon if its color is green, tastes sweet, and round.
- These features might be dependant on each others, however, we assume they are all independent and that's why its 'Naive'!



PRACTICE OPPORTUNITY

NAÏVE BAYES: QUIZ/CALCULATE THE PROBABILTY OT NON-RETIRING (RED

$$P(No\ Retire|X) = ?$$

NAÏVE BAYES: QUIZ/CALCULATE THE PROBABILTY OT NON-RETIRING (RED CLASS)

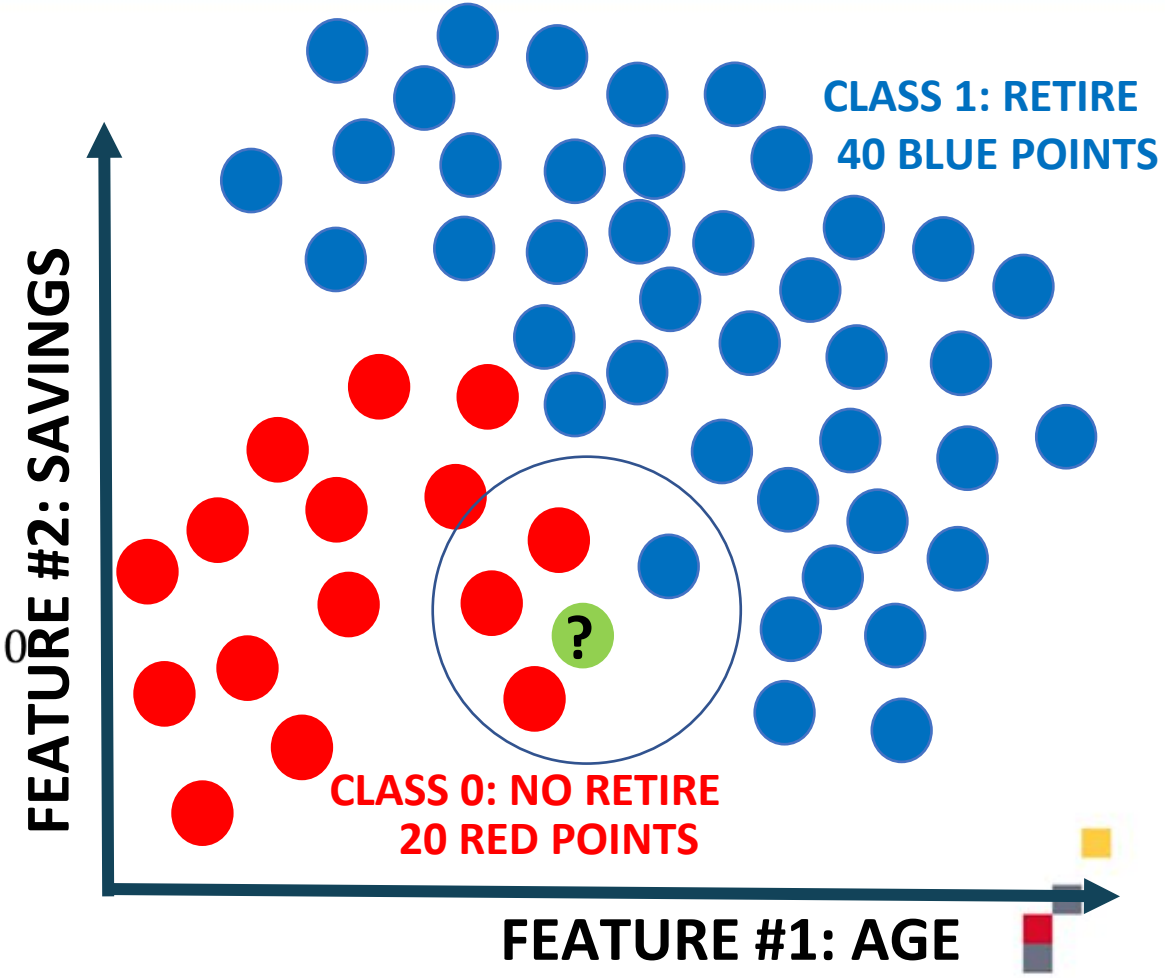
LIKELIHOOD

PRIOR PROBABILITY OF NO RETIRING

$$P(No\ Retire|X) = \frac{P(X|No\ Retire) * P(No\ Retire)}{P(X)}$$

MARGINAL LIKELIHOOD

- $P(No\ Retire) = \frac{\text{\# of No Retiring}}{\text{Total points}} = 20/60$
- $P(X|No\ Retire) = \frac{\text{\# of smilar observations for No retiring}}{Total\ \#\ no\ retiring} = 3/20$
- $P(X) = \frac{\text{\# of Similar observations}}{Total\ \#\ Points} = 4/60$
- $P(No\ Retire|X) = \frac{\frac{20}{60} * \frac{3}{20}}{\frac{4}{60}} = \frac{3/60}{4/60} = 0.75$



$$\boldsymbol{NOTE: P(Non\ Retire|X) = 1 - 0.25 = 0.75}$$